

Can a Local Model Beat the Frontier?

Ontology-Grounded DiffusionGemma vs. Bare Cloud Models

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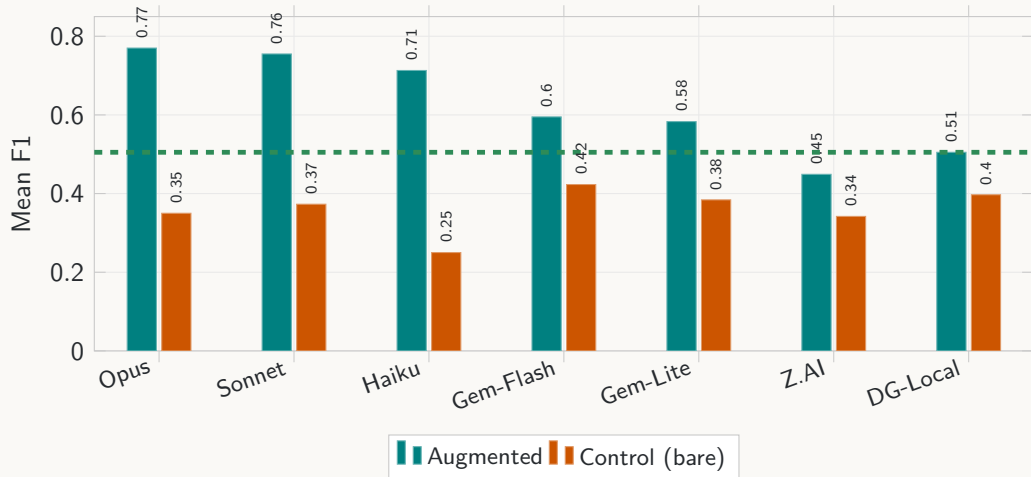
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The Question

Can a **local model** running on one GPU,
grounded in a **formal knowledge graph**,
match **frontier cloud models** on domain recall?

7 models ■ 672 isolated runs ■ 16 KG-oracle questions ■ deterministic grading

Headline: Local + Ontology Beats Every Frontier Bare



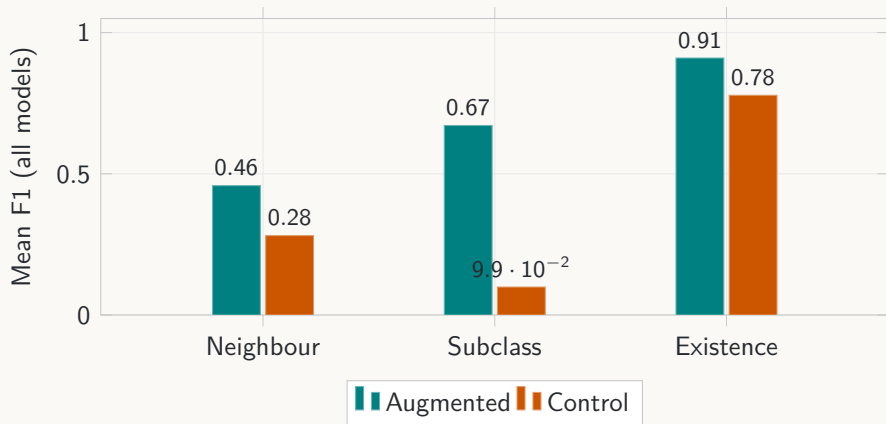
Dashed line = local DiffusionGemma + ontology. Every frontier bare score falls below it.

The Numbers

Model	AUG F1	Bare F1	Δ	Halluc drop
Opus 4.8	0.770	0.350	+0.420	-0.529
Sonnet 4.6	0.755	0.373	+0.382	-0.480
Haiku 4.5	0.713	0.250	+0.463	-0.608
Gemini 3.5 Flash	0.595	0.423	+0.172	-0.210
Gemini 2.5 Flash Lite	0.583	0.384	+0.199	-0.232
Z.AI (GLM-4.6)	0.449	0.342	+0.108	-0.206
DiffusionGemma (local)	0.505	0.397	+0.108	-0.199

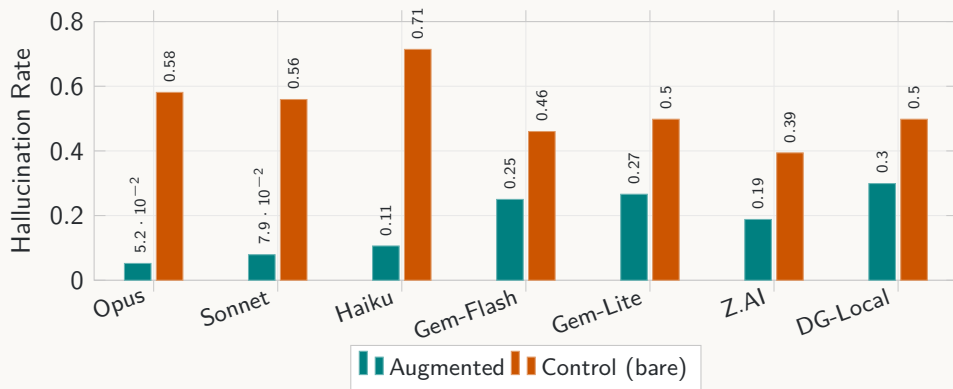
Every model benefits. Hallucination drops across the board.

Where Grounding Helps Most



Subclass: $6.8\times$ improvement. Models have near-zero parametric knowledge of ontology hierarchies.

Hallucination Halves with Grounding



Overall: $0.529 \rightarrow 0.177$. KG grounding provides verifiable facts, not plausible guesses.

Cost, Privacy, and Data Sovereignty

	DG Local + Ontology	Frontier AUG	Frontier Bare
F1 (best)	0.505	0.770 (Opus)	0.423 (Gem-Flash)
Data leaves org?	No	Yes	Yes
Per-query cost	\$0*	\$0.01–0.08	\$0.002–0.04
Latency	2–4 s	15–50 s	5–15 s
GDPR-compliant?	Yes	Depends	Depends

For regulated industries:

- ▶ Zero data exfiltration
- ▶ Predictable, flat cost
- ▶ Sub-second KG lookup

Ontology condense cost:

- ▶ One-off: 6h for 7,445 classes
- ▶ Re-index only when KG changes
- ▶ Per-query: sub-ms cache read

*Amortised electricity/hardware only.

Important Caveats

1. **Not like-for-like.** Local model receives KG triples; frontier models get only the question. If frontier models were also grounded, they score much higher (Opus AUG: 0.770).
2. **Frontier models gain more from grounding.** Claude +0.38–0.46 vs. DiffusionGemma +0.108. The ontology is a *capability multiplier*—it scales with the base model.
3. **DG's own lift is modest.** +0.108 F1. Its strength is a surprisingly competitive bare score (0.397), not extraordinary use of the ontology.
4. **Performance varies by type.** DG excels on existence (0.875) but struggles on neighbour recall (0.323) due to the 256-token block decode limit.
5. **Domain-specific only.** Results measure KG fidelity, not general QA.

Key Takeaways

1. **Ontology grounding works** — universally, across 7 models
+0.265 F1, hallucination halved
2. **Local + ontology > frontier bare** — for domain recall
DiffusionGemma 0.505 beats Opus bare 0.350
3. **Data sovereignty is achievable** — without sacrificing accuracy
Zero API cost, zero data exfiltration, GDPR-ready

672 runs ▪ 7 models ▪ 7,445 OWL classes ▪ deterministic grading

DreamLab AI ▪ Dr John O'Hare ▪ ontology-augment (PRD-020)